## Practice questions on User-defined functions and Views

**Q1A. Create a view named "Salesman_Sales_Total" that displays the total sales made by each salesman.**

```
CREATE VIEW Salesman_Sales_Total AS
SELECT s.salesman_id, s.name AS salesman_name, SUM(o.amount) AS total_sales
FROM Salesman s
INNER JOIN Orders o ON s.salesman_id = o.salesman_id
GROUP BY s.salesman_id;
```

**Q1B. Create a user-defined function that takes sales_id as the input and returns the total sales of the salesperson.**

```
DELIMITER $$
CREATE FUNCTION salestotal
(s_id INT)
RETURNS DECIMAL(10,2)
DETERMINISTIC
BEGIN
   DECLARE tot_sales DECIMAL(10,2);
   SELECT total_sales into tot_sales FROM Salesman_Sales_Total
   WHERE salesman_id = s_id;
   RETURN tot_sales;
END $$
DELIMITER ;
```

**Q1C. Find the sales total sales done by the salesman with id no. 106.**

```
mysql> Select salesman_id, name, salestotal(salesman_id) from salesman where
salesman_id = 106;
```

**Q2A. Create a view with only customer id, customer name and level from customers table.**

```
mysql> create view level as select customer_id, c_name, level from customers;
```

**Q2B. Write a user-defined function that takes the customer id as the input and returns the letter grade based on their level using the above view. If the level is 100, the grade is ‘C’, if the level is 200, the grade is ‘B’ and if the level is 300, the grade is ‘A’.**

```
DELIMITER $$
CREATE FUNCTION levelGrade(c_id INT)
RETURNS CHAR
DETERMINISTIC
BEGIN
   DECLARE grade char;
   DECLARE tempLevel INT;

   SELECT level into tempLevel FROM level WHERE customer_id = c_id;

   If tempLevel = 100 then
      SET grade = 'C';
   END IF;

   If tempLevel = 200 then
      SET grade = 'B';
   END IF;

   If tempLevel = 300 then
      SET grade = 'A';
   END IF;

   RETURN grade;
END $$
DELIMITER ;
```

**Q2B) Display the grade of the customer with id 504.**

```
Select levelGrade(504);
```